

Dataset inspection report – Employee attrition analysis

1. Introduction

The dataset used for this analysis is the IBM HR Analytics – Employee Attrition & Performance dataset. Before carrying out any detailed exploratory analysis, an initial inspection was conducted to understand the structure, characteristics, and overall quality of the data. This process involved examining the number of records and variables, data types, missing values, duplicate records, and descriptive statistics. The purpose of this initial assessment was to determine whether the dataset was suitable for further analysis and to identify any data quality issues that might need to be addressed.

2. Dataset structure

The dataset consists of 1,470 employee records and 35 variables. Each row represents an individual employee, while each column contains information about a particular employee characteristic or workplace related factor. The variables cover a range of areas, including employee demographics, job roles, income, job satisfaction, working conditions, length of employment, and employee attrition. Examples of the variables included in the dataset are Age, Department, JobRole, MonthlyIncome, JobSatisfaction, OverTime, YearsAtCompany, and Attrition. The Attrition variable is particularly important for this analysis because it indicates whether an employee left the organization (Yes) or remained with the organization (No). This variable will therefore be used as the main outcome when investigating factors that may be associated with employee turnover.

3. Data types

The dataset contains a combination of numerical and categorical variables, allowing the data to be analyzed from different perspectives. Numerical variables are primarily stored as integer (int64) values. Examples include Age, DistanceFromHome, MonthlyIncome, TotalWorkingYears, and YearsAtCompany. These variables can be used to examine numerical patterns and differences among employees. Categorical variables are stored as string (str) values and include variables such as Attrition, Department, Gender, JobRole, MaritalStatus, and OverTime. These variables are useful for comparing employee groups and examining whether particular categories are associated with higher or lower levels of attrition.

4. Missing value assessment

A missing value assessment was performed using the Pandas `isnull().sum()` function. The results showed that there were no missing values across any of the 35 variables. This means that all 1,470 employee records contain information about the variables included in the dataset. As a result, no missing value treatment or imputation was necessary during the initial data cleaning stage.

5. Duplicate record assessment

The dataset was also checked for duplicate records using the Pandas `duplicated().sum()` function. The result was zero, indicating that no completely duplicated rows were identified. The absence of duplicate records is important because it reduces the possibility of individual employee records being unintentionally counted more than once. This provides greater confidence that each row represents a unique employee record for the purposes of the analysis.

6. Descriptive statistics

Descriptive statistics were examined for the numerical variables to obtain an initial understanding of the characteristics of the workforce. The average employee age was 36.92 years, with ages ranging from 18 to 60 years. The median age was 36 years, indicating that the typical employee in the dataset was in their mid-thirties. Employees had an average of approximately 11.28 total working years, while the average number of years spent with the organization was approximately 7.01 years. The number of years employees had been with the organization ranged from 0 to 40 years, showing considerable variation in employee tenure. The average distance from home was 9.19, with values ranging from 1 to 29. This variable may be worth examining further because the distance an employee travels to work could potentially influence job satisfaction, work-life balance, and ultimately employee retention. Two variables, `EmployeeCount` and `StandardHours`, showed no variation across the dataset. `EmployeeCount` had a constant value of 1, while `StandardHours` had a constant value of 80 for all employees. Since these variables have the same value for every record, they do not provide meaningful differences between employees and are therefore unlikely to contribute useful information to the exploratory analysis.

7. Categorical variable summary

The categorical variable assessment showed that the dataset contains two possible outcomes for employee attrition: Yes and No. Of the 1,470 employees, 1,233 employees had an Attrition value of No, while 237 employees had an Attrition value of Yes. This represents an overall employee attrition rate of approximately 16.12%, while approximately 83.88% of employees remained with the organization. This attrition rate provides an important baseline against which differences between employee groups can be examined during the subsequent analysis. Research & Development was the most represented department, with 961 employees. In terms of business travel, Travel_Rarely was the most common category, accounting for 1,043 employees. The most frequently occurring job role was Sales Executive, with 326 employees, while the most common overtime status was No, with 1,054 employees. The Over18 variable contained only one category, Y, across all 1,470 employee records. Since there is no variation in this variable, it is unlikely to provide useful information when comparing employees or investigating factors associated with attrition.

8. Overall data quality assessment

The initial inspection indicates that the dataset is of suitable quality for further exploratory analysis. It contains 1,470 employee records across 35 variables, with no missing values and no duplicate records. The combination of numerical and categorical variables also provides sufficient information to investigate different characteristics that may be associated with employee attrition. The overall attrition rate of 16.12% provides a useful baseline for the next stage of the analysis. Further exploratory analysis will investigate whether factors such as department, age, monthly income, overtime, and job role are associated with differences in employee attrition. Overall, the initial dataset inspection provides a clear understanding of the structure and quality of the data and establishes a suitable foundation for the exploratory analysis that follows.